

# Supplementary Material

## Controlled Component Comparisons and Cross-Cohort Failure Analysis for Single-Channel EEG Sleep Staging

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This supplement contains implementation details and descriptive tables moved from the main manuscript. The evidence labels are unchanged: seed 42 is primary, seed 123 is a post-protocol sensitivity repeat, SHHS1 E1/E2 analyses are secondary on an opened cohort, and SHHS1 E3–E2/E4 analyses are post-hoc. E6 estimates mean and standard deviation from each complete unlabelled target recording; it is label-free but transductive at recording level, not purely inductive zero-shot inference.

### Supplementary Methods S1: model and training details

The control encoder contains 15 CNN branches: three context groups (current, previous and next; C/P/N) crossed with five stage-specific detectors. Each branch uses two convolutional blocks and a small head; their outputs form a 75-dimensional feature vector. The ResNet-1D encoder uses a convolutional stem, three residual blocks and global average pooling to produce a 128-dimensional representation from the current epoch only. Thus, E1–E0 isolates the sequence-model substitution, whereas E2–E1 replaces the encoder, removes explicit P/N context and changes feature dimension.

The TCN projects inputs to 128 dimensions and uses six residual blocks with dilations 1, 2, 4, 8, 16 and 32, symmetric padding and dropout 0.2, giving a 253-epoch receptive field. The BiLSTM has 128 hidden units per direction. Encoders are fitted first, selected on validation data, hash-verified and used to cache features without gradients; sequence models are then fitted on these frozen caches. CNN and BiLSTM optimisation uses Adam; ResNet-1D uses AdamW; TCN uses Adam. Exact learning rates are  $10^{-3}$ ,  $10^{-2}$ ,  $10^{-3}$  and  $5 \times 10^{-4}$ , respectively. Batch sizes are 64 epochs for encoders, four recordings for BiLSTM and eight recordings for TCN. No label smoothing or data augmentation is used.

### Supplementary results

Table 1: S1. Per-class Sleep-EDF metrics for E3 and E6 (seed 42).

| Stage | E3 Prec. | E3 Rec. | E3 F1 | E6 Prec. | E6 Rec. | E6 F1 | $\Delta F1$ |
|-------|----------|---------|-------|----------|---------|-------|-------------|
| W     | .9338    | .9271   | .9304 | .9391    | .9169   | .9279 | +0.0026     |
| N1    | .5149    | .5410   | .5276 | .5032    | .5219   | .5124 | +0.0152     |
| N2    | .8583    | .8447   | .8514 | .8401    | .8368   | .8385 | +0.0130     |
| N3    | .8072    | .8097   | .8085 | .7160    | .7779   | .7457 | +0.0628     |
| REM   | .8274    | .8413   | .8343 | .8227    | .8197   | .8212 | +0.0131     |

Table 2: S2. Pooled E3 confusion matrix on Sleep-EDF (rows: reference; columns: prediction).

|     | W      | N1     | N2     | N3     | REM    |
|-----|--------|--------|--------|--------|--------|
| W   | 61,133 | 3,888  | 416    | 56     | 448    |
| N1  | 3,283  | 11,643 | 4,867  | 105    | 1,624  |
| N2  | 539    | 5,496  | 58,394 | 2,271  | 2,432  |
| N3  | 30     | 45     | 2,377  | 10,558 | 29     |
| REM | 483    | 1,542  | 1,984  | 90     | 21,736 |

Table 3: S3. Sleep-EDF macro-F1 under two fixed seeds.

| Configuration | Seed 42 | Seed 123 | Difference |
|---------------|---------|----------|------------|
| E0            | .7754   | .7755    | +.0000     |
| E1            | .7802   | .7776    | -.0026     |
| E2            | .7835   | .7828    | -.0007     |
| E3            | .7904   | .7883    | -.0022     |
| E4            | .7891   | .7887    | -.0004     |
| E6            | .7691   | .7780    | +.0089     |

Table 4: S4. Seed sensitivity of pre-specified Sleep-EDF contrasts; Holm correction is within seed.

| Contrast | Seed 42: $\Delta$ [95% CI], $p_{\text{Holm}}$ | Seed 123: $\Delta$ [95% CI], $p_{\text{Holm}}$ |
|----------|-----------------------------------------------|------------------------------------------------|
| E1–E0    | .004811 [–.000915, .010193], .1022            | .002188 [–.002870, .007043], .9130             |
| E2–E1    | .003251 [–.002370, .008808], .1036            | .005143 [–.000571, .011292], .2400             |
| E3–E2    | .006962 [.000305, .014520], .8989             | .005478 [–.002611, .015796], .9130             |
| E3–E6    | .021319 [.012179, .030698], .0012             | .010249 [.002435, .017989], .1313              |

Table 5: S5. Seed-123 wall-clock training and validation time (10 folds, sequential, same GPU).

| Configuration                | Total            | Mean per fold   |
|------------------------------|------------------|-----------------|
| E0: 15-CNN + BiLSTM          | 33 h 35 min 36 s | 3 h 21 min 34 s |
| E1: 15-CNN + TCN             | 14 min 17 s      | 1 min 26 s      |
| E2: ResNet-1D + TCN          | 2 h 44 min 57 s  | 16 min 30 s     |
| E3: fixed scale              | 3 h 16 min 40 s  | 19 min 40 s     |
| E4: band-pass                | 2 h 55 min 56 s  | 17 min 36 s     |
| E6: target-record $z$ -score | 2 h 24 min 44 s  | 14 min 28 s     |

Table 6: S6. Seed-123 sensitivity on the same 180 SHHS1 participants.

| ID | Subject-mean macro-F1 | Pooled macro-F1 | Accuracy | $\kappa$ |
|----|-----------------------|-----------------|----------|----------|
| E0 | .5303                 | .5785           | .6738    | .5420    |
| E2 | .5409                 | .5858           | .6874    | .5597    |
| E3 | .5634                 | .6031           | .7003    | .5772    |
| E4 | .5732                 | .6147           | .7064    | .5869    |
| E6 | .5503                 | .5865           | .6873    | .5552    |

Table 7: S7. Per-class transfer of E3; Pred/True is a marginal emission ratio, not calibration.

| Stage | Sleep-EDF |           |       | SHHS1 |       |           | $\Delta F1$ |
|-------|-----------|-----------|-------|-------|-------|-----------|-------------|
|       | F1        | Pred/True | Prec. | Rec.  | F1    | Pred/True |             |
| W     | .9304     | .993      | .8823 | .8014 | .8399 | .908      | -.0905      |
| N1    | .5276     | 1.051     | .2528 | .5437 | .3451 | 2.151     | -.1824      |
| N2    | .8514     | .984      | .7394 | .7349 | .7372 | .994      | -.1143      |
| N3    | .8085     | 1.003     | .9440 | .2582 | .4055 | .273      | -.4030      |
| REM   | .8343     | 1.017     | .6052 | .8939 | .7218 | 1.477     | -.1125      |

Table 8: S8. Pooled E3 confusion matrix on SHHS1 (rows: reference; columns: prediction).

|     | W      | N1    | N2     | N3    | REM    |
|-----|--------|-------|--------|-------|--------|
| W   | 29,638 | 4,481 | 1,171  | 21    | 1,672  |
| N1  | 874    | 3,807 | 601    | 3     | 1,717  |
| N2  | 2,348  | 5,947 | 56,063 | 324   | 11,605 |
| N3  | 78     | 39    | 16,674 | 5,888 | 127    |
| REM | 655    | 784   | 1,311  | 1     | 23,183 |

Table 9: S9. Per-class SHHS1 metrics for E3 and E2.

| Stage | E3 Prec. | E3 Rec. | E3 F1 | E2 Prec. | E2 Rec. | E2 F1 | $\Delta F1$ |
|-------|----------|---------|-------|----------|---------|-------|-------------|
| W     | .8823    | .8014   | .8399 | .8780    | .7275   | .7957 | +.0442      |
| N1    | .2528    | .5437   | .3451 | .2367    | .3528   | .2833 | +.0618      |
| N2    | .7394    | .7349   | .7372 | .7362    | .6937   | .7143 | +.0228      |
| N3    | .9440    | .2582   | .4055 | .9659    | .2496   | .3967 | +.0087      |
| REM   | .6052    | .8939   | .7218 | .4886    | .9450   | .6442 | +.0776      |

Table 10: S10. Oracle prioritisation on E3 predictions. Values are potential leverage, not achievable performance.

| Condition            | macro-F1 | Numerical gap recovered |
|----------------------|----------|-------------------------|
| Observed E3 on SHHS1 | .6099    | —                       |
| Correct N3→N2 only   | .7444    | .1345 (74.5%)           |
| Correct N2→REM only  | .6596    | .0497 (27.5%)           |
| E3 on Sleep-EDF      | .7904    | —                       |

Table 11: S11. Post-hoc reference-label boundary and stable-interior diagnostic (seed 42).

| Cohort    | ID | Boundary macro-F1 | Stable macro-F1 | Boundary N3 recall | Stable N3 recall |
|-----------|----|-------------------|-----------------|--------------------|------------------|
| Sleep-EDF | E0 | .6179             | .8175           | .6121              | .9216            |
| Sleep-EDF | E3 | .6321             | .8346           | .6326              | .9316            |
| SHHS1     | E0 | .4323             | .5959           | .0633              | .4008            |
| SHHS1     | E3 | .4689             | .6220           | .0733              | .3900            |

E6 N3 recall is 0.2005 overall and 0.0721 in the same SHHS1 boundary neighbourhood, compared with 0.2582 and 0.0733 for E3. These regions use reference labels on both sides and are offline diagnostics, not physiological transitions or deployable detectors.

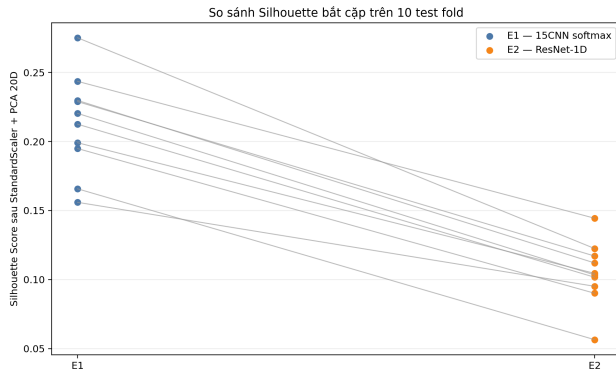
## Supplementary analyses: representation and context

The mean silhouette coefficient is 0.2126 (SD 0.0356) for E1 and 0.1048 (SD 0.0229) for E2. This comparison is descriptive: E1 consists of 75 class-posterior features, whereas E2 is an unconstrained 128-dimensional representation, so silhouette is not a like-for-like measure of downstream utility.

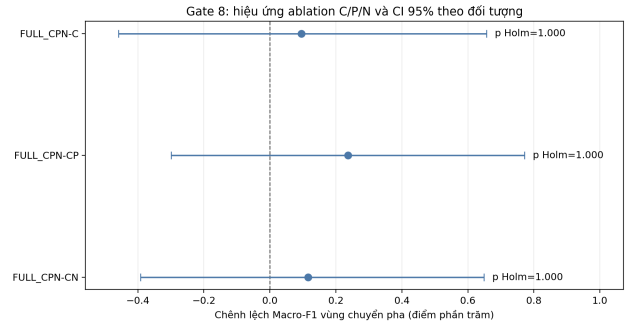
Table 12: S12. Context-group ablation within E1.

| Condition | Overall macro-F1 | $\Delta$ vs C | Boundary macro-F1 | $\Delta$ vs C | N1 F1 |
|-----------|------------------|---------------|-------------------|---------------|-------|
| C         | .777477          | —             | .619102           | —             | .5011 |
| CP        | .777733          | +0.000257     | .617686           | -.001416      | .5022 |
| CN        | .781585          | +0.004108     | .618895           | -.000207      | .5055 |
| CPN       | .780230          | +0.002753     | .620055           | +0.000953     | .5115 |

For the pre-specified CPN–C boundary comparison,  $\Delta_{\text{macro-F1}} = 0.000953$ , 95% CI  $[-0.004588, 0.006568]$  and Holm  $p = 1.000$  (36/78 subjects improved). Secondary CPN–CP and CPN–CN intervals also include zero with Holm  $p = 1.000$ . The result does not show that temporal context or P/N information is useless; it only finds no detectable marginal benefit for this encoding in E1.



(a) Silhouette by fold.



(b) Context-ablation effects.

Figure 1: S1. Supporting representation and context analyses.

## Provenance limitation

Prediction artifacts and recomputed metrics are hash-verified. The SHHS run-manifest protocol hash ('165d7cdf...fe93') matches the retained historical snapshot `configs/shhs_zero_shot_v1.json`; the currently checked-out `configs/shhs_v1_protocol.json` ('9541e233...fe9') is an expanded post-run audit record, not a replacement. The reconciliation is recorded in `Reports/SHHS_PROTOCOL_PROVENANCE.md`. This resolves the snapshot-level provenance discrepancy, while raw SHHS data and prediction artifacts remain outside the repository and E6 remains a descriptive reanalysis of locked predictions.